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12 Dec 1962

MILITARY SPECIFICATION
BOOSTER, AMMUNITION, ELECTRIC, CALIBER .50

This specification is approved for use by all Departments and Agencies of the Department of Defense.

SCOPE

1.1 Scope. This specification covers all electric ammunition boosters for the caliber .50, M3 machine gun.

1.2 Classification. Electric ammunition boosters for the caliber .50, M3 machine gun shall be one of the following types and classes (see 6.2):

Type I - Equipped with D.C. electric motor.

Type II - Equipped with A.C. electric motor.

Class 1 - The booster will drive counterclockwise when viewed from the wire end and the projectile end sprockets will be next to the wire end.

Class 2 - The booster will drive counterclockwise when viewed from the wire end and the primer end sprockets will be next to the wire end.

Class 3 - The booster will drive clockwise when viewed from the wire end and the projectile end sprockets will be next to the wire end.

Class 4 - The booster will drive clockwise when viewed from the wire end and the primer end sprockets will be next to the wire end.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletion) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMEDTB, Robins AFB, GA 31096-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. document or by letter.

AMSC N/A FSC 1005

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2.1.1 Specifications and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specification and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

MIL-P116	Preservation, Methods of
MIL-M-7969	Motor, Alternating Current, 400-Cycle, 115/200-Volt System, Aircraft General Specification for
MIL-M-8609	Motor, DC, 28 Volt System, Aircraft, General Specification for

STANDARDS

Military

MIL-STD-129	Marking for shipment and storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-769	Part and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Method and Engineering Guidelines

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subject to first article inspection and testing (see 4.4 and 6.2).

3.2 Design. The ammunition booster assembly shall be designed for installation in ammunition boxed or elsewhere in the fire control system to aid the conveying of ammunition from the ammunition box to the feedway of the gun.

3.3 Construction. The ammunition booster shall be a compact unit so constructed that no parts will work loose in service. It shall withstand the strains, jars, vibrations and other conditions incident to shipping, storage, and service. The motor wires shall extend from the end opposite the anti-roll-back release.

3.4 Maintainability. The ammunition booster shall be constructed to allow replacement of component parts with ease during overhaul or servicing.

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3.5 Reliability. The operational life of the unit shall be a minimum of 84,000 rounds of ammunition passing over the booster when operated on a duty cycle (see 4.3).

3.6 Performance. The ammunition booster shall meet the performance requirements under the operating conditions as called our in Section 4.

3.7 Components. The ammunition booster shall consist of the following: an electric motor, two sprocket wheels, freewheeling clutch and a anti-roll-back device assembled together within fixed end plates.

3.7.1 Motor.

Type I - The electric motor required shall be in accordance with the applicable requirements of MIL-M-8609 for intermittent duty motors.

Type II - The booster shall be driven by a 200 volt, 3 phase, 400 cycle motor in accordance with MIL-M-7969 except as modified herein.

3.7.2 Electrical properties.

Type I - The booster at room temperature will not exceed a maximum of 16 amperes at 26 volts d. c. across the motor terminals when pulling 1400 rounds of ammunition per minute under a load of 50 pounds.

Type II - The booster at room temperature will not exceed a maximum of 2 amperes at 200 volts, 400 cycles, three phase alternating current across the motor terminals when pulling 1400 rounds of ammunition per minute under a load of 50 pounds.

3.7.3 Sprockets. The booster sprocket wheels shall be arranged and placed as shown (see Figure 1).

3.7.4 Freewheeling clutch. The ammunition booster shall be constructed to allow the "housing assembly" to free wheel in the direction of drive with no indication of binding or seizing due to deformed bearings, soft bearing surfaces (raceways), mismatched parts or faulty workmanship.

3.7.5 Anti-roll-back device. The assembly will contain an anti-roll-back device to prevent booster rotation in a direction opposite to feeding rotation. This device will be spring loaded when normally connected and will be mechanically disconnectable for reverse rotation.

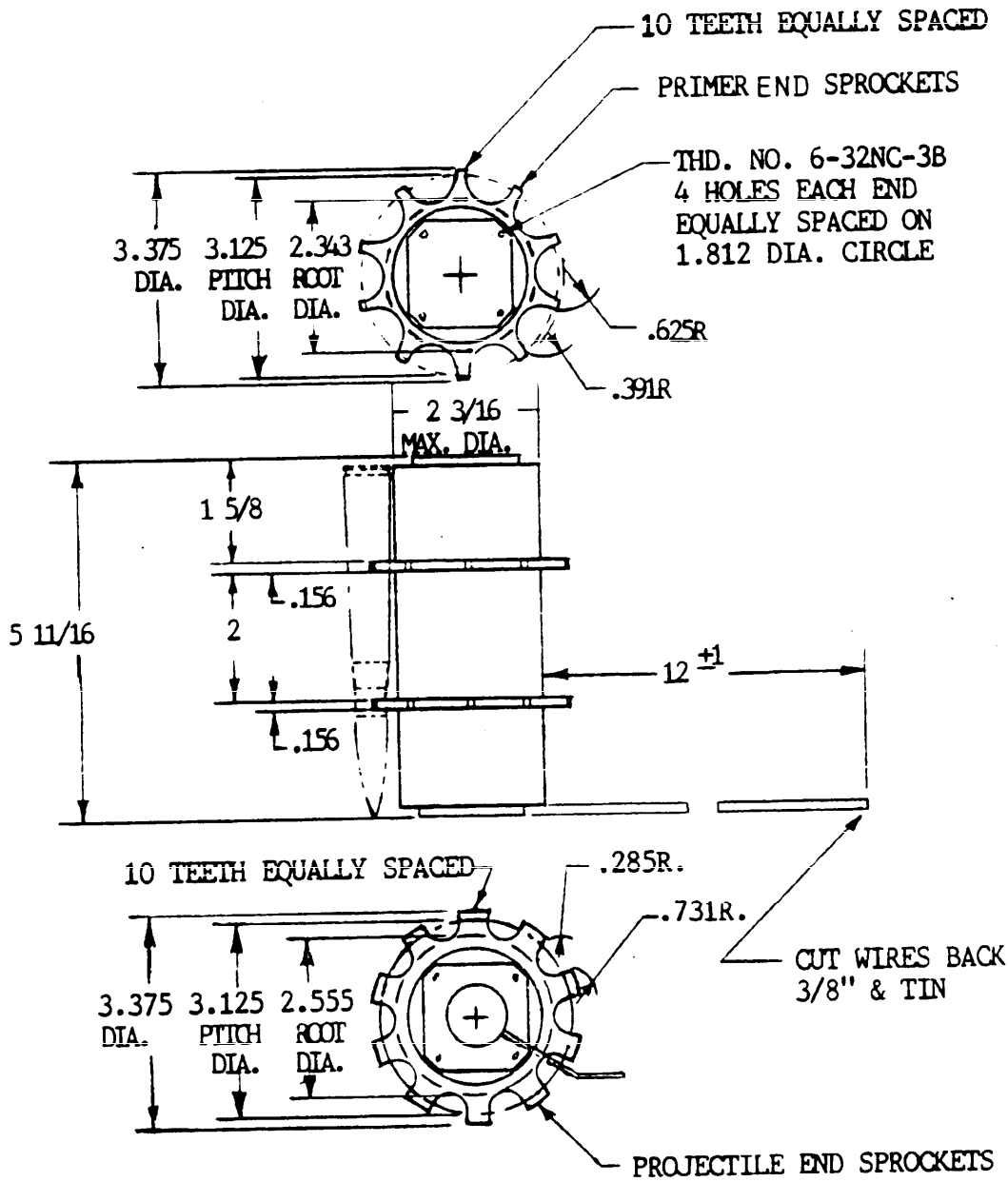
3.8 Dimensions. Dimensions for the ammunition booster shall be as shown (see Figure 1).

3.9 Weight. Weight of the ammunition booster shall not exceed five pounds.

3.10 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130 using the following information:

Booster, Ammunition, Electric

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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCE ON:
-ANGLES-
-DECIMALS—±.005
-FRACTIONS—±.1/64

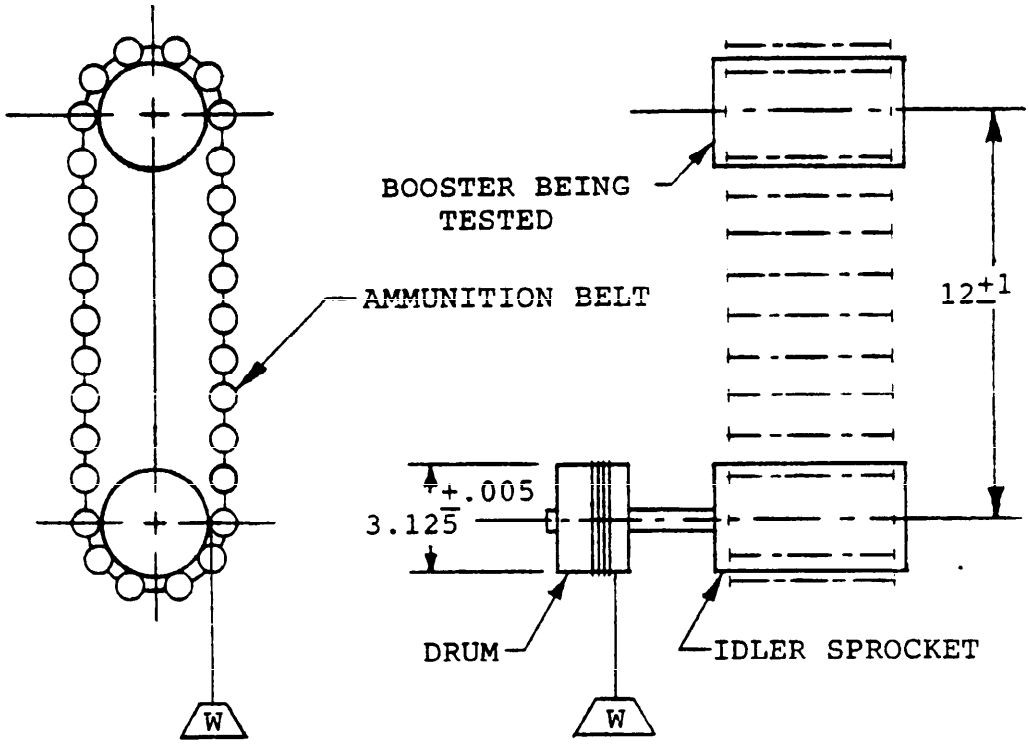
BOOSTER-AMMUNITION
ELECTRIC, 26V D.C/200V A.C
CALIBER .50

FIGURE 1

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THE WEIGHT OF THE IDLER SPROCKET DRUM AND CABLE SHALL BE AS LIGHT AS POSSIBLE TO REDUCE THE INERTIA LOAD

PITCH DIA. OF BOOSTER AND IDLER SPROCKET 3.125

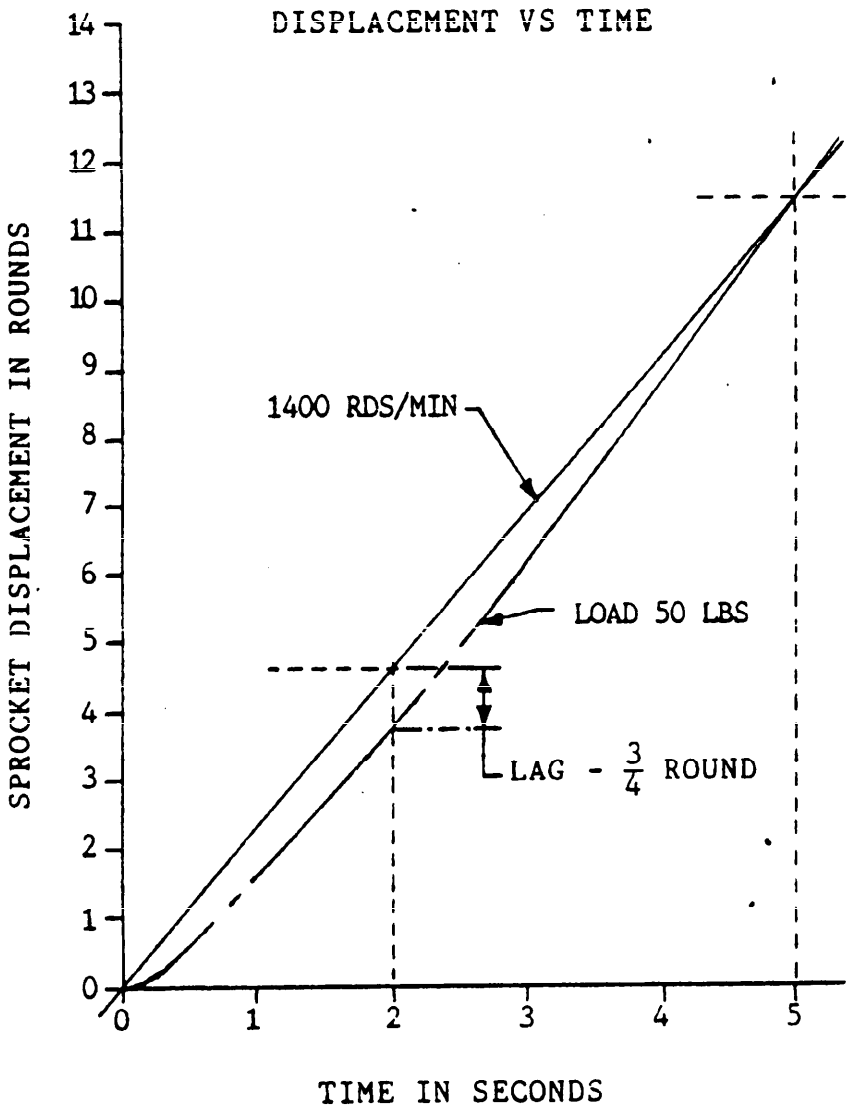


BOOSTER TEST RIG

FIGURE 2

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ove all loose dirt and foreign matter work on outer side of cooling coil. Remo



GRAPH OF BOOSTER ACCELERATION WITH 50 LB LOAD

FIGURE 3

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Caliber .50, M3 Machine Gun

Type Voltage System _____

Specification No.

Mfr's Part No.

Mfr's Serial No.

Order No.

Mfr's Name or Trademark

U.S. Property

3.11 Workmanship. All detail of workmanship shall be of high quality to ensure satisfactory operation and service life. Parts shall not contain sharp edges, burrs, loose chips, dirt or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective materials.

4.2 Classification of inspections. The inspection and testing of booster shall be classified as follows:

a. First Article Inspection (see 4.3).

b. Quality Conformance Inspection (4.4).

4.3. Preproduction inspection and test. A sample of each type of booster (see 1.2) the supplier furnishes under this contract shall be inspected and tested as follows:

4.3.1 Examination of product. The preproduction sample shall be thoroughly

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examined to assure conformance to all requirements of this specification. Characteristics not covered by this specification shall be examined to assure conformance to the manufacturer's specification.

4.3.2 Duty cycle. The duty cycle shall be four seconds on, four seconds off, for four minutes of each hour.

4.3.3 Preproduction test. The preproduction tests shall consist of the following tests in the order they are given and those specified under inspection test method, subject to the test conditions specified herein. All tests are to be conducted.

4.3.4 Cycle rate at room temperature. The steady state of cycle rate with a 50 pound load shall be a minimum of 1400 rounds per minute when tested with a setup similar to the one shown (see Figure 2).

After a 72 hour soak at -65°F and at an ambient of -65°F the booster cycle rate at 50 pound load shall be a minimum of 1200 rounds per minute throughout four duty cycles. After the completion of the -65°F tests, the booster shall be subjected to -100°F for at least one-half hour and at an ambient of -100°F the cycle rate shall be at least 1200 rpm with 50 pound load for one duty cycle.

4.3.5 Stall characteristics - Type I. The sample shall be tested to determine that it will withstand 25 stalls of four seconds' duration spaced five minutes a part with no loss in efficiency when cooled at room temperature with power source of 28 volts d.c. across the motor terminals. The stalls may be accomplished by any applicable means.

4.3.6 Stall characteristics - Type II. The sample shall be tested to determine it will withstand 25 of four seconds' duration, space five minutes apart with no loss in efficiency when cooled at room temperature with power source of 200 volts, 400 cycles, alternating current across the motor terminals. The stalls may be accomplished by any applicable means.

4.3.7 Load versus acceleration test. The sample shall be tested to determine its acceleration under rated load. The booster acceleration under a 50 pound load shall be such that in .20 seconds a cycle rate of 1400 rounds per minute shall be attained, at which time the booster shall not trail the gun fire rate (1400 rounds per minutes) by more than 0.75 rounds. Further, it shall have caught up with the gun displacement within 0.50 seconds. Graphically the requirement shall be as shown in Figure 3.

4.3.8 Anti-roll-back test. The anti-roll-back device shall be able to withstand a 120 pound load at the pitch diameter. The maximum backlash with a load of 45 pounds on the sprocket pitch diameter shall not exceed three degrees between the sprocket and the booster mounting frame. With the anti-roll-back in the release position, the booster sprocket shall roll back with a force not to exceed 6 pounds applied at the pitch diameter.

4.3.9 Environmental tests. These tests shall consist of those specified below of MIL-STD-810:

Tests	Method NO.
Humidity	507.2

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Altitude	520.0
Salt Spray	509.2
Vibration	514.3
Fungus Resistance	508.3
Sand and Dust	510.2
Explosion Proof	511.2

4.3.10 Life test. The second sample shall be tested to determine capability of performing intermittent duty of four seconds' "on", four seconds "off" for a period of four minutes of each hour under a load of 50 pounds and at a rate of 1400 rounds per minute through a minimum of 3600 seconds operation, equivalent to 84,000 rounds of ammunition passing over the booster.

4.3.11 Freewheeling clutch test. The sample used in life test shall freewheel in the direction of drive when a force not to exceed 1 pound is applied to the pitch diameter.

4.4 Acceptance inspection. Acceptance inspection of each unit shall include all examinations necessary to insure compliance with the requirements of the specifications listed herein, and shall include tests contained in 4.3.

4.4.1 Inspection of preparation for delivery requirements. Packing, packaging, and preservation shall be in accordance with Section 5.

5. PACKAGING

5.1 Preservation and packaging. Unless otherwise specified, preservation and packaging shall be level A and shall be accomplished in accordance with Method 11 B of MIL-P-116. Unit package quantity shall be one each.

5.2 Packing. Packing shall be level A or B, as specified (see 6.2) and shall be accomplished in accordance with MIL-STD-794, except that metal shipping that metal shipping containers shall not be used.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The type of ammunition booster covered by this specification is intended to be used to convey or assist in the flow of ammunition from the box to the caliber .50 machine gun.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and number of preproduction samples, point of testing and delivery of preproduction test report.

c. Selection of applicable levels of packing.

6.3 Subject term key word listing.

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a. Booster, ammunition.

b. Caliber .50.

c. Military specification.

Custodian
Air Force - 99
Army - AR

Preparing Activity
Air Force - 84
Agent: Air Force - 99
Project 1005-0702

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	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

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b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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